AIDAN GETTEMY | MATHEMATICS

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PROFILE

Ph.D. candidate in Mathematics at the University of Texas at Dallas (advisor: Dr. Felipe Pereira), graduating February 2027. My research develops computational tools for high-dimensional uncertainty quantification, with a focus on Bayesian inverse problems, MCMC methods, and surrogate modeling, applied to fluid flow in porous media and wind energy. I have experience with large real-world datasets (SCADA), high-dimensional simulation output, and parallel computing on HPC systems.

EDUCATION

Doctor of Philosophy in Mathematics Graduation: Feb. 2027
University of Texas at Dallas, Richardson, TX
Dissertation: "*Uncertainty quantification methods for subsurface flows and wind energy*"
Advisor: Felipe Pereira

Bachelor of Music Performance and Mathematics May 2021
Baylor University, Waco, TX
Summa Cum Laude, Honors

Technical Skills

Analysis Tools: Bayesian inverse problems; Markov chain Monte Carlo; uncertainty quantification; surrogate modeling; numerical partial differential equations.
Programming: Python; MATLAB; C; R; Julia; Bash; LaTeX.
Software: Git; Linux; SLURM; MPI; Jupyter; VS Code; RStudio; Anaconda; MATLAB Parallel Computing Toolbox; scikit-learn; NumPy; Paraview; OpenFAST.
Data Management: Parquet; SQL; large-scale simulation output management.
Techniques: Finite difference methods; Finite volume methods; Gaussian random field modeling; Gaussian process regression; domain decomposition methods for PDEs.

RESEARCH EXPERIENCE

Research Assistant, University of Texas at Dallas, Richardson, TX Aug. 2024 - Aug. 2026

- Developed global-local multiscale MCMC frameworks for Bayesian inverse problems in domain-decomposed Darcy flow models, improving computational efficiency.
- Investigated prior-preserving proposal mechanisms for Gaussian random fields parameterized by Karhunen-Loève expansions.
- Evaluated MCMC convergence using R-hat, coverage, reconstruction error, and posterior fields.
- Developed and maintained MATLAB/Python code and ran large-scale experiments on university HPC systems (SLURM, MPI, bash), managing simulation outputs and reproducible experiment records across projects.
- Presented results at conferences and in weekly meetings with international collaborators, helped onboard junior researchers, and contributed to manuscript drafts.

Engineering Intern, Roux Institute, Portland, ME June - Aug. 2024

- Built Python SCADA data pipelines (pandas, MATLAB, Parquet) and filtering and visualization tools for scalable wind farm performance analysis.
- Studied wake-steering bias and turbine-to-turbine flow interactions in utility-scale SCADA data, using regression to identify yaw misalignment biases and improve alignment of steered and unsteered datasets.
- Developed consensus, random forest, and Gaussian process models to estimate unobserved turbine states and reconstruct spatial wind fields.
- Collaborated with Roux Institute and WindESCo teams, presented at a research conference, and contributed major analysis to a journal manuscript now under review.

Graduate Researcher, University of Texas at Dallas, Richardson, TX Aug. 2023 - Aug. 2024
Mentors: Susan Minkoff and John Zweck.

- Simulated wind turbine behavior under varying inflow conditions and leading-edge erosion severity using OpenFAST to support condition-monitoring model development
- Built a MATLAB framework to automate simulation runs, data processing, and analysis of OpenFAST outputs.

- Evaluated supervised learning and Gaussian process emulator models for erosion damage detection, contributing analysis, figures, and manuscript text to the published Wind Energy Science article.

Computational Research, University of Texas at Dallas, Richardson, TX June - Aug. 2022

Mentor: Michael Zhang.

- Studied spin-glass models and computational approaches for biological system identification with the UTD Computational Biology Lab.
- Mentored an undergraduate student researcher on project context, modeling concepts, and research workflow.

Publications

Journal Papers

Gettemy, A., Minkoff, S., Zweck, J., and Spiller, E., "Classification of Leading-Edge Erosion Severity via Machine Learning Surrogate Models," Wind Energy Science. *Published* September 10, 2026, https://doi.org/10.5194/wes-11-3377-2026.

Journal Papers in Review

Gettemy, A., Abbatessa, L., and Post, N., "Evaluating Yawed Turbine Transfer Functions from SCADA Data," *In review*: Preprint available in Wind Energy Science.

Journal Papers in Preparation

Gettemy, A., Borges, M., Guiraldello, R., and Pereira, F., "Global-Local Anchored Multiscale Method for Inverse Modeling of High-dimensional PDEs." *In preparation*

Academic Projects

Karhunen-Loève Expansion and Darcy Flow Adjoint Problem,

Personal Project July 2026

- Derived the algorithm to compute the gradient of pressure-error function with respect to the KLE coefficients using finite volumes and adjoint formulation.
- Implemented the finite volume adjoint in MATLAB, and tested on optimization using MATLAB Optimization Toolbox and tested Bayesian inversion version using Hamiltonian Monte Carlo in MATLAB Statistics and Machine Learning Toolbox, verifying usefulness of gradient information.

Portfolio Optimization and Risk Management Project,

Class: MATH *Stochastic Dynamic Programming* April 2025

- Formulated and solved portfolio optimization problems using Markowitz mean-variance theory to analyze the risk-return tradeoff in investment allocation.
- Implemented optimization models with MOSEK and presented results demonstrating how constraints and risk preferences affect optimal portfolio selection.

Bayesian Inverse Methods for Infectious Disease Forecasting,

Class: STAT *Bayesian Data Analysis* Dec. 2024

- Implemented Bayesian compartmental growth models for epidemic forecasting, using MCMC methods to estimate uncertain model parameters from observed case data.
- Conducted R/C++ analysis and rolling-origin cross-validation to evaluate predictive performance across sequential forecast horizons.

Comparing Machine Learning approaches for Wind Turbine Blade Damage Detection,

Class: PHYS *Big Data and Machine Learning for Scientific Discovery* April 2024

- Applied MATLAB machine learning tools to model wind turbine erosion patterns using supervised regression and classification methods.
- Compared neural networks, decision trees, linear models, discriminant analysis, Naive Bayes, random forests, and error-correcting output code classifiers, identifying random forest as the best-performing approach after hyperparameter tuning.

Multi-scale method C coding project,

Class: MATH *Scientific Computing* Aug. - Dec. 2023

- Developed a finite-difference solver for elliptic PDEs in C for modeling Darcy flow in porous media.
- Implemented MPI-based domain decomposition algorithm to partition the computational grid across processors and accelerate PDE solves on HPC systems.
- Evaluated solver performance by comparing computational speed, accuracy, and scalability across serial and parallel implementations.

Hurricane Database Construction June 2021

Personal Project

- Built a relational SQL database for NOAA hurricane records, supporting structured analysis of historical storm data and developed Python workflows for data extraction, cleaning, API-based collection, and exploratory visualization.

Presentations

Mini-symposium Presentation, "Progress on the Global-Local Method for High-Dimensional Inverse Problems," UTD Comet Computing Conference (CCC) 2026, Sep. 16, 2026, Richardson, TX.

Campus Seminar Presentation, "Multi-scale MCMC Methods for PDE-based Bayesian Inverse Problems", UTD STAT Exchange Graduate Seminar, April 17, 2026, Richardson, TX.

Campus Seminar Presentation, "A Global-Local Multi-scale MCMC Method for PDE-based Bayesian Inverse Problems", UTD Computational Science Seminar, March 26, 2026, Richardson, TX.

Mini-symposium Presentation, "A Global–Local Multiscale MCMC Method for Domain-Decomposed Bayesian Inference", SIAM Conference on Uncertainty Quantification (UQ26), March 22-25, 2026, Minneapolis, MN.

Poster Presentation, "Global–Local Multiscale MCMC for Domain-Decomposed Bayesian Inference", CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning, Dec. 8-12, 2025, Houston, TX.

Mini-symposium Presentation, "Evaluating Yawed Turbine Transfer Functions from SCADA Data," NAWEA/WindTech 2025, Oct. 15-17, 2025, Richardson, TX.

Mini-symposium Presentation, "Fast converging Markov chain Monte Carlo methods for high dimensional inverse problems," SIAM Texas-Louisiana Section Annual Meeting, Sep. 26-28, 2025, Austin, TX.

Mini-symposium Presentation, "Fast converging Markov chain Monte Carlo methods for high dimensional inverse problems," UTD Comet Computing Conference (CCC) 2025, Sep. 17, 2025, Richardson, TX.

Poster Presentation, "Domain decomposed Bayesian inference," ChAMELEON Summer School, Aug. 11-15, 2025, Houston, TX.

Mini-symposium Presentation, "Multi-scale MCMC Sampling," SIAM Conference on Computational Science and Engineering (CSE25), March 3-7, 2025, Fort Worth, TX.

Mini-symposium Presentation, "A Digital Twin Approach for Monitoring Wind Turbine Erosion," SIAM Texas-Louisiana Section Annual Meeting, Oct. 11-13, 2024, Waco, TX.

Research Symposium Presentation, "Digital Twins for Wind Energy Applications," 3D 4D Seismic Consortium Yearly Research Meeting Presentation, May 10, 2024, Richardson, TX.

Campus Seminar Presentation, "Introduction to the Sparse Identification of Nonlinear Dynamics," UTD Computational Science Seminar, Oct. 14, 2022, Richardson, TX.

Honors and Awards

Student Travel Award, CBMS Conference Dec. 2025
AAML Workshop Student Travel Award, University of Houston, Houston, TX

Dean's Award for Outstanding Student in Music Performance April 2020
Baylor University, Waco, TX

Membership in Pi Kappa Lambda Music Honor Society April 2020
Baylor University, Waco TX

Round Top Festival Cellist June - July 2019
Round Top Festival Institute, Round Top, TX

University Dean's List Dec. 2017 - May 2021
Baylor University, Waco, TX

President's Gold Academic Scholarship Aug. 2017 - May 2021
Baylor University, Waco, TX

Getterman Music Scholarship Aug. 2017 - May 2021
Baylor University, Waco, TX

Professional Training

Applied Mathematics and Machine Learning CBMS Conference Dec. 8 - 12, 2025
University of Houston, Houston, TX

ChAMELEON Summer School Aug. 11 - 15, 2025
University of Houston, Houston, TX
University of Houston Graduate Courses Aug. - Dec. 2021
University of Houston, Houston, TX
Introduction to Scientific Computing June - Aug. 2021
University of Washington, Seattle, WA

TEACHING EXPERIENCE

Teaching Assistant, Natural Sciences and Mathematics Jan. 2022 - May 2024 and Aug. - Dec. 2026
University of Texas at Dallas, Richardson, TX
Across appointments: ran problem sessions, held weekly office hours, graded, and supported students through Canvas and Blackboard. Completed the Graduate Reflective Teaching Seminar and departmental TA training.

- Fall 2026: MATH 2414 Integral Calculus (assisted; two sections of about 20) and MATH 6313 Numerical Analysis (graded; about 20 graduate students)
- Aug. 2022–May 2024: MATH 3310 Theoretical Concepts of Calculus (graded; about 80) and MATH 3311 Abstract Algebra I (graded; about 40)
- Spring 2022: MATH 2419 Calculus II, Honors (assisted; led a weekly two-hour problem session, wrote study guides, coordinated exams, graded)

Grading Assistant, Baylor University, Waco, TX Aug. 2020-May 2021
- Graded undergraduate mathematics assignments in combinatorics, algebra, and linear algebra, providing consistent evaluation and feedback as an undergraduate music major.
- Managed assignment materials and grading records using Canvas LMS, spreadsheets, and Adobe Acrobat to support course organization and instructor workflows.

EMPLOYMENT HISTORY (MUSIC RELATED)

Contractor, KATY ISD, Katy, TX Oct. 2021 - Jan. 2022
- Led a middle school clinic for students preparing for Texas Music Educators Association Region 23 honor orchestra and adjudicated high school orchestra auditions.
- Taught private cello lessons at Haskett Junior High School.

Contractor, Music & Arts, Sugar Land, TX Oct. 2021 - Jan. 2022
- Provided private cello instruction for beginning and intermediate students.
- Designed individualized lesson plans and documented student progress to support consistent technical growth and prepare future lessons.

Intern/Cello Performer, Waco Symphony, Waco, TX Sep. 2017 - May 2021
- Prepared and performed orchestral repertoire for approximately three concerts per semester, while supporting ensemble rehearsals and community outreach performances.

Teacher, Central Texas String Academy, Waco, TX Sep. 2019 - May 2021
- Taught private and group cello lessons using Suzuki-based principles, developing individualized exercises, lesson materials, and practice notes to support student progress.
- Supported student recitals during rehearsal preparation and concert.

SERVICE

Service to Professional Organizations

Symposium Chair Mar. 24, 2026
SIAM Conference on Uncertainty Quantification 2026, C5-Surrogate Models
Symposium Co-Organizer Sep. 26, 2025
SIAM Texas-Louisiana Section Annual Meeting
Mini-symposium Title: “Domain Decomposition for Coupling and Inverse Problems”
SIAM Student Volunteer, SIAM CSE, Fort Worth, TX Mar. 4, 2025
- Helped SIAM employees manage the desk: late registration, badges, and provide directions to rooms.

Symposium Co-Organizer Oct. 11, 2024
SIAM Texas-Louisiana Section Annual Meeting October 2024
Mini-symposium Title: “Applications and Methods for Digital Twins”

Service to Department

Workshop Co-Organizer Mar. 2026

UTD-SMU SIAM Student Chapter, *Introduction to Variational Auto-Encoders*

- Co-organized a technical workshop introducing variational autoencoders to graduate students and early-career researchers.
- Coordinated event logistics, including room reservation, advertising, outreach, and communication with student participants.

Workshop Co-Organizer Feb. 2026

UTD-SMU SIAM Student Chapter, *Physics Informed Neural Networks using MATLAB*

- Communicated with multiple stakeholders, at MathWorks, SIAM Student Chapter, and the UTD MathWorks ambassador.
- Raised awareness of the event, advertised through email channels, and helped to arrange logistics.

Conference Volunteer Mar. 2023

University of Texas at Dallas

Conference: "Advances in Stat. and Comp. Methods for Analysis of Biomedical, Genetic, and Omics Data"

- Supported conference operations by preparing event spaces, setting up poster-session areas, posting directional signage, and guiding attendees to reception locations.
- Contributed to conference side-programming by performing cello duets before banquet reception.

Conference Volunteer Feb. 2022

University of Texas at Dallas

Conference: "Texas Geometry and Topology Conference"

- Assisted with conference operations, including room setup, poster-session preparation, visitor directions, and transportation support for invited guests.

Service to University

University of Texas at Dallas SIAM Student Chapter Sep. 2023 - Sep. 2025

Co-President, Richardson, TX

- Presided over meetings, worked with sponsoring faculty members to organize colloquia, helped organize food logistics and transportation to meetings, and promoted events to increase public awareness.
- Organized multiple events per year including; presentations from Department of Energy Lab Scientists, resume building sessions, interviews with mathematical alumni, and a museum field trip.

Service to Community

University of Texas at Dallas Musica Nova Chamber Ensemble Mar. 2022 - May 2026

Musician, Richardson, TX

- Performed in semesterly chamber ensemble concerts, contributing to campus arts programming through weekly rehearsals and public performances.

Professional Memberships

Society of Industrial and Applied Mathematics (SIAM) Sep. 2022 - present

LANGUAGES

English: Native Language

CITIZENSHIP

USA: Citizen

HOBBIES AND INTERESTS

Cello, Running, Hiking, Birding, Plants, Art, Reading